Yang-Jun Joo, Ph.D.

Data Scientist | AV Performance Evaluation & Statistical Modeling

Autonomous Vehicle Safety | Measurement Frameworks | Large-Scale Mobility Data

Orlando, FL | Open to relocation to Mountain View / San Francisco

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Professional Summary: Applied data scientist with 8+ years of quantitative research and applied data science experience, including 3+ years post-Ph.D. Developed AV performance evaluation and measurement frameworks, new safety metrics, and Python/SQL analytics pipelines for large-scale on-road and mobility data. Expertise in rare-event rate estimation, combining real and synthetic data, experimental design, anomaly investigation, and uncertainty-aware evaluation of large-scale ML models.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, scikit-learn, PyTorch, deep learning, Git, Linux, ETL pipelines, data visualization.
Statistics & ML Evaluation:	Probabilistic modeling, Bayesian networks, reliability analysis, causal inference, experimental design, rare-event rate estimation, real and synthetic data integration, uncertainty quantification, model calibration.
AV & Mobility Analytics:	AV performance evaluation, traffic modeling, safety evaluation and prediction, trajectory prediction, on-road/naturalistic driving, connected-vehicle event data.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar, Quantitative Safety Analytics University of Central Florida Mar 2024-Present

- Collaborated across agency and technical teams to develop evaluation frameworks and standardized safety-performance metrics using connected-vehicle, naturalistic-driving, and high-resolution event data.
- Built Python/SQL ETL and analytics pipelines for multi-terabyte data from 1,000+ intersections, enabling standardized metric generation, trend interpretation, and anomaly investigation for operational decisions.
- Designed and analyzed near-miss risk-judgment experiments across 926 participants and 1,325 crash/near-miss clips, applying causal forests to quantify effects of vehicle automation.
- Developed synthetic data generation methods for class-imbalanced crash severity estimation, supporting safety prediction under sparse-event conditions.
- Evaluated large-scale driving-scene VLMs with conformal prediction, identifying context-specific calibration failures for safety-critical model evaluation.

Postdoctoral Researcher (AV Safety Systems) Seoul National University Mar 2023-Feb 2024

- Led research on real-time infrastructure-to-vehicle (V2I) safety information systems for the Korean National Police Agency, defining safety-critical scenarios and operational safety criteria.

Graduate Research Assistant Seoul National University Mar 2018-Feb 2023

- Developed new risk-field and reliability-based metrics for AV lane-change and collision-avoidance evaluation using trajectory prediction, Bayesian networks, and surrogate safety measures.

SELECTED PUBLICATIONS

- Kim, Kim, and Joo* (2026), "Generative AI for Class Imbalance in Crash Severity Estimation with Mixed Data Types," Transportation Research Record.
- Joo et al. (2023), "A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory," Analytic Methods in Accident Research.
- Joo et al. (2021), "Reliability-Based Assessment of Potential Risk for Lane-Changing Maneuvers," Transportation Research Record.

EDUCATION

Ph.D., Civil & Environmental Engineering (Transportation), Seoul National University, 2023

B.S., Civil & Environmental Engineering, Seoul National University, 2018